Assignment #2: Tangible and Embodied UI

- **Submission:** a four-page written report using the ACM template (single column), submitted on Fenix by the end of Week 7 (Wednesday, October 21, until 23:59) via Fenix as a PDF. References do not count towards page limits
- **Challenge:** design and implement a tangible and embodied interface for the first assignment
- **Expected result:** a functional prototype that allows the user to control the vehicle from the first assignment using physical (Arduino) and/or gestural controls (Femto Bolt)
- **Assessment** (0-20v): 6v for adherence to the theoretical concepts covered in class; 6v for cyberphysical integration (i.e., does the interface work as intended?); 5v for how you respond to the peer critique session in Week 6; and 3v for the quality of the written report (contents and presentation)

1. Challenge

The goal of your team in this assignment is to **design and implement a tangible and embodied interface** for assignment #1. For that, you will support physical controls using the Arduino microcontroller (Figure 2) and various sensors (light, sound, movement, temperature, etc.) and, optionally, support gestures via computer vision using Orbbec's Femto Bolt depth camera. The goal is not to emulate existing controllers, but to design and implement **a novel and intuitive user interface that leverages several of the concepts introduced in our theoretical classes**: from *coupling and phenomenology* (Paul Dourish), to *material semantics* (Brygg Ullmer), to *physical affordances* (Eva Hornecker), to *image schemas* (Jörn Hurtienne).

2. Planning

The (strongly) suggested work plan for the Labs in Weeks 4-7 is shown in Figure 1.

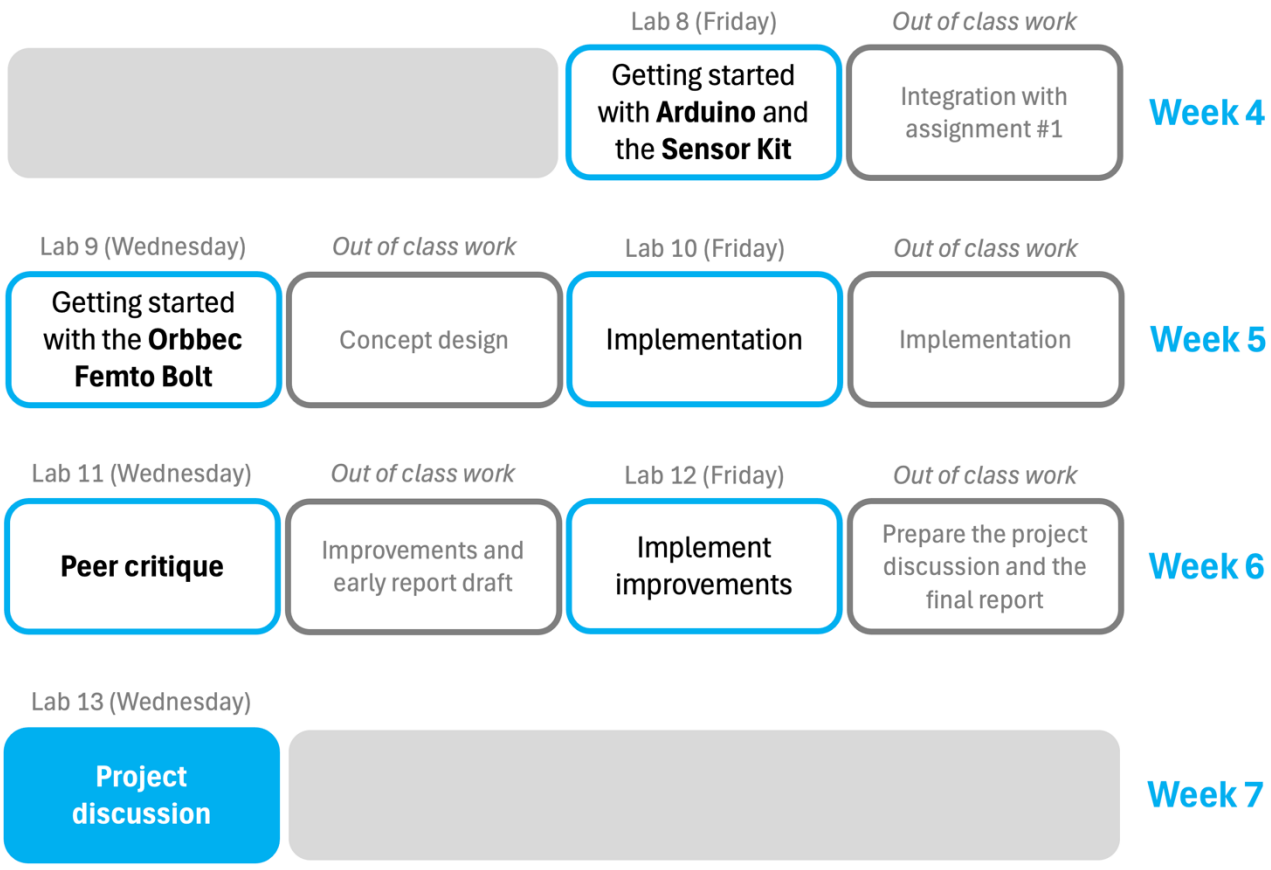


Figure 1. The suggested work plan for the labs in Weeks 4-7, including recommended activities between labs.

2. Recommendations

- **Labs in Weeks 4-6 are working spaces where you continue to *learn how to learn***. That is, the role of the lab instructor is not to demonstrate how every potential sensor works nor to troubleshoot your code, but to point to appropriate resources and discuss your ideas. Arduino is a well-documented, easy-to-use platform aimed at a wide range of educational levels. In addition to the resources below in Section 3, you are free to learn via AI virtual assistants and other online resources.
- **We do not assess code quality, but the rationale behind your interface and interaction design (and how well it works).** It is critical that you leverage the concepts introduced in the theoretical classes. You should complement these classes with the recommended readings listed in Section 3.3 below.
- **Less can be more.** Projects that leverage more sensors will not necessarily have higher marks. While we value ambition, as we mentioned above, the assessment centres around the rationale behind your interface and interaction design.
- Consider that **your interface is not necessarily the sensors but the physical enclosure around them.** For instance, if your interface uses a physical button, instead of having users interact with Arduino's tiny button, you should build an appropriately sized button that presses this instead. To do this, you can 3D print or laser-cut your case at Lisbon's FabLab, repurpose old electronics, or use materials such as foamcore, Plastimake, modelling clay (e.g., Fimo), Sugru, and ballast materials like steel washers to add weight to your interface (if needed).

3. Resources and Recommended Readings

3.1 Arduino

- Getting started with the Arduino
- Arduino Sensor Kit tutorials
- Arduino Sensor Kit open-source library
- PlatformIO IDE for VSCode

3.2 Orbbec Femto Bolt

- Getting started with the Orbbec Femto Bolt
- Orbbec SDK and K4A Wrapper
- Getting started with the Google MediaPipe Hands

3.3 Recommended Readings

- Where the Action Is: The Foundations of Embodied Interaction, *Paul Dourish*
- Weaving Fire into Form: Aspirations for Tangible and Embodied Interaction, *Brygg Ullmer et al.*
- Getting a grip on tangible interaction: a framework on physical space and social interaction, *Eva Hornecker et al.*
- Image schemas and their metaphorical extensions: intuitive patterns for tangible interaction, *Jörn Hurtienne et al.*
- How bodies matter: five themes for interaction design, *Scott R. Klemmer et al.*

4. Submission

Students should submit a four-page report by the end of Week 7 (Wednesday, October 21, until 23:59) via Fenix. This should document their design and rationale (with appropriate references), their implementation and any iterations following the peer critique in Week 6, supporting figures, links to online repositories and 3D models (if 3D printing was used), etc. You need to use the ACM template (*single column*) and submit it as a PDF file. References do not count towards the page limit, and only one student per group needs to complete the Fenix submission. To generate a single-column layout using the template above, use the following command: `\documentclass[manuscript, screen, review]{acmart}`.

A select number of projects will be submitted to the 21st ACM Tangible, Embedded, and Embodied Interaction Conference (TEI '27), taking place in Lisbon at Centro Cultural de Belém from January 24-27, 2027. We will cover the registration costs of students accepted to the conference so that they can present their work in person. ACM is the Association for Computing Machinery, and being accepted to TEI means your report was peer-reviewed by academics and researchers in the field and is part of the conference proceedings (i.e., it has a DOI and you are now a published author). This type of output is particularly important to students looking to apply for scholarships to national and international PhD programmes. To improve your chances of success, please have a look at the theme of this year's Student Design Competition (and example past projects).

5. Assessment

- **Theoretical adherence [6v].** Does your interface perfectly map to one or more of the theoretical pillars discussed in class (and in the resources above in Section 3.3)? Is it novel and intuitive?
- **Cyberphysical integration and execution [6v].** Do the sensors work as intended with imperceptible latency or jitter, and the physical enclosure is robust and withstands repeated use?
- **Response to the peer critique session in Week 6 [5v].** Did you clearly identify interaction breakdowns highlighted in the critique and significantly improve your interface and interaction design?
- **Quality of the written report [3v].** Does your report successfully present and ground your concept design and implementation work? Is it supported by appropriate figures and references? Is it a suitable submission to the Student Design Competition at ACM TEI '27?



Figure 2. The Arduino microcontroller and the Arduino Sensor Kit with a button, potentiometer, accelerometer, and light, sound, air pressure, and temperature sensors. Each student group is provided one of these sets at the start of Week 5, which must be returned undamaged to the lab instructor at the end of the course.